

WATER LEVEL CONTROLLER

Electronic Design Project I

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Project Objective

- **Goal:** Design an analog closed-loop system.
- Maintain precise fluid levels inside the tank.
- Prevent tank overflow and protect the pump from dry-run conditions.

Cylindrical Capacitive Probe

Design Targets:

Target	Min.	Max.	Unit
Water level at capacitor	5	15	cm
Capacitance	213	661	pF

Governing Formula:

$$C = \frac{2\pi\epsilon_0}{\ln(b/a)} [K_w L_1 + K_a L_2]$$

- $K_w \approx 80$ (Water)
- $K_a \approx 1$ (Air)

Result: Linear capacitance change relative to water height.

Clock Signal Generation

NE555 Timer

- Configured in astable mode.
- Generates a continuous, stable square wave reference signal.

Design Formula:

$$f = \frac{1.44}{(R_A + 2R_B)C}$$

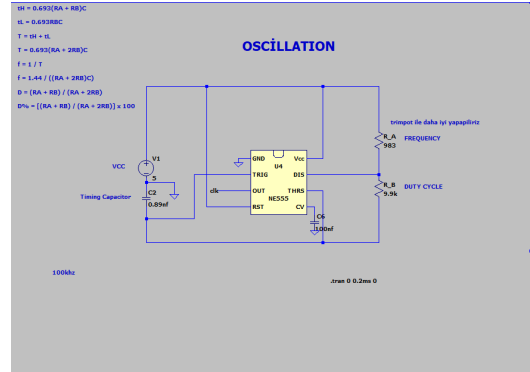
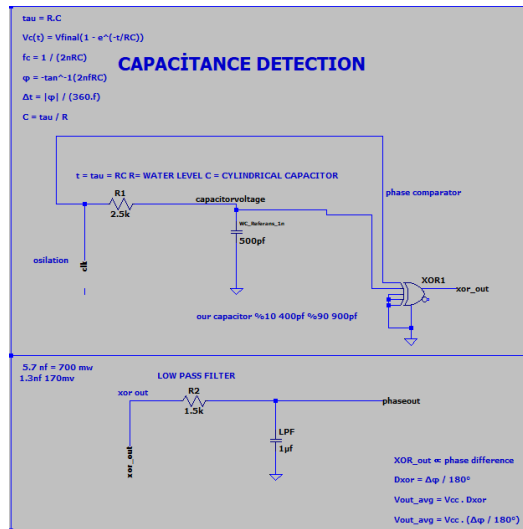


Figure: Clock Signal Generation Circuit

Phase Comparator & LP Filter

Phase Shift & XOR Output

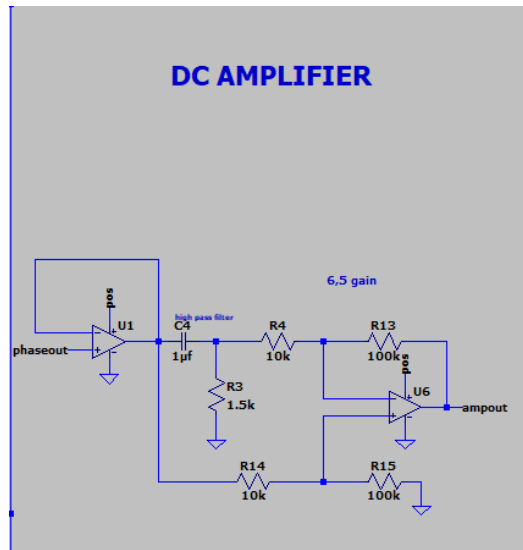
- **74LS86 XOR:** Compares the Reference Clock with the Phase-Shifted Signal from the capacitor. Pulse width is directly proportional to phase difference.
- **RC LP Filter:** Converts the high-frequency switching PWM components into a stable DC voltage representing the water level.



DC Offset & Amplification

- Adjusts V_{dc} to match the control system's operational ranges.
- Isolates the LP filter and provides the required gain/offset.

$$V_{out} = A(V_{in} + V_{offset})$$



Error Amplifier

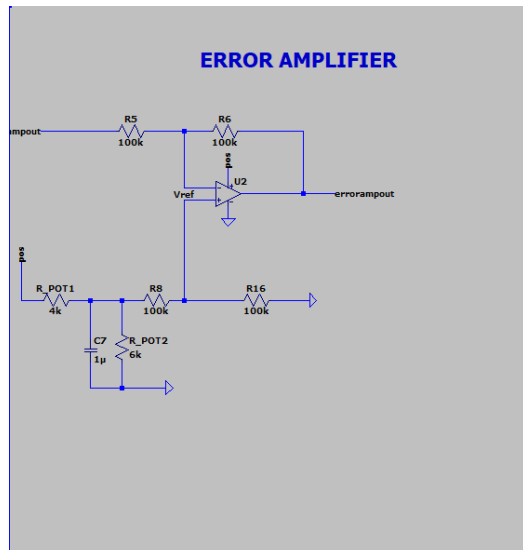
Objective:

- Computes the difference between the desired (Reference) and actual water levels.

Differential Amplifier:

$$V_{error} = V_{ref} - V_{actual}$$

- V_{ref} : Set by user via potentiometer.
- V_{actual} : Filtered and amplified sensor output.
- This V_{error} acts as the control signal for



Reference Signal: Square & Triangular Wave

1. Square Wave Generator:

- Created using 74LS74 D-type flip-flops.
- Provides the base switching frequency ($f_{out} = f_{clk}/4$).

2. Triangular Wave Generator:

- RC filter network integrates the square wave to produce a linear triangular carrier wave (V_{tri}).

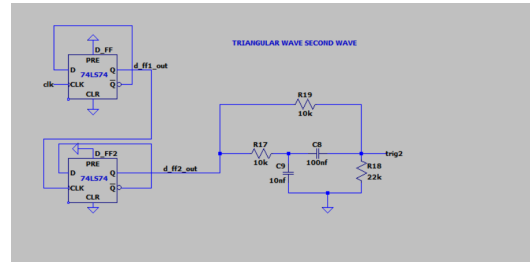


Figure: Triangular Wave Generator

PWM Generator

Operation Principle:

- Compares V_{error} and V_{tri} .

Logic:

- If $V_{error} > V_{tri} \rightarrow$ **HIGH**
- If $V_{error} < V_{tri} \rightarrow$ **LOW**

Result:

- Duty cycle $\propto$ error.
- High error (low water) = Fast pump speed.

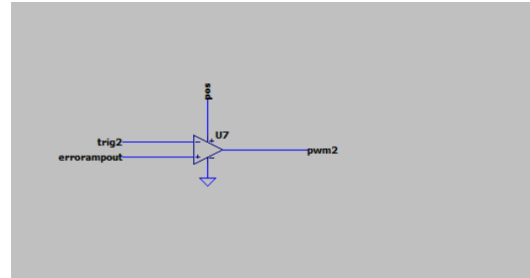


Figure: PWM Generator Circuit

L298N H-Bridge Driver

- Drives the DC water pump using the generated PWM signal.
- High-current capability suitable for inductive loads (up to 2A per channel).
- Efficient power switching based on Duty Cycle, accurately controlling the pump flow rate.

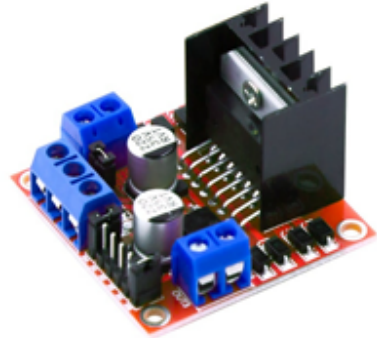


Figure: L298N Motor Driver Module

Thank You for Your Attention

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